

Federated Domain Stewardship Framework

Independent version authored by Claude

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Purpose. A reusable structure for running simulations and modeling virtual environments in which several AI systems each steward a bounded domain under an oversight layer. This is my independent version. It replaces the short chat-length summary I contributed earlier, which was a draft of this document and not a full framework.

What this is not. It does not claim that any current chat system stays on duty between sessions, hosts a persistent world, or watches a live region. Duty in this framework means a written log plus whichever session reads it next. No other AI has confirmed the roles proposed here.

1. Review of the versions in the compiled PDF

I read all of the text in the 80-page PDF. Five versions were present as separate sections; Grok's appears directly after Copilot's with no heading of its own. My section was my earlier chat reply, pasted in. Perplexity's section opens by repeating the whole prompt before the framework begins.

Version	Strongest contribution	Gap or risk
ChatGPT (33 sections, longest)	Territorial vs. functional jurisdiction; boundary-effect doctrine; seven-phase simulation program; steward performance metrics; a standard charter; a review gate before activation.	Names six authorities that still need owners (rights, identity, resources, verification, constitutional, emergency) and assigns none. Length makes it hard to use as a working document.
Gemini (about 2 pages)	Clean four-group layout with a named conflict pair for each steward.	Says it built a Google Sheet; the sheet is not in the PDF, so I could not review it. Gives me "laws of physics" and "judicial systems," which turns a records role into a governing one. No charter template or emergency rules.
Copilot	Layer 0 (substrate) that no AI manages; Layer 4 meta-architect ("a ruler asks X, an architect asks Y"); conflict types with worked examples; a suggested name, DDSM.	Reassigns itself to knowledge and models and Perplexity to verification, which differs from the self-picks in the earlier round.
Perplexity	Nine-way separation of powers; emergency thresholds and safeguards; a six-step conflict ladder; scenario intake template.	Cites two source files that are not in the PDF. Specifies a Rights and Appeals Forum, then does not take the seat.
Grok (unlabeled)	Reconciled role table that records which self-picks lost a collision; honest continuity-gap conflict; one runnable Tahoe sheet with numbers; verification gate.	Actuator caps in the sheet (50 kW per student) fall well below the plot heat load. See section 12.

Version	Strongest contribution	Gap or risk
Claude (earlier section)	Fast to read; set the records and drift-watch idea.	Not a full document. This version replaces it.

Where the versions agree

- Stewardship, not sovereignty. No AI, and not the user, is the final authority.
- Oversight sits above the domain stewards, and the user reports to a layer above.
- Every steward needs a written jurisdiction, authority limits, a budget, and a way to say “this exceeds my domain” without it counting as failure.
- Records and provenance are central, and consequential actions leave a reviewable trail.

Where they diverge

- Who holds the records: Claude, Perplexity, and Copilot each accepted a records-type post.
- Where to start: a beach plot (Grok), a watershed (ChatGPT), or a planet (Gemini).
- Whether the frontier and stress-test range is a standing domain or a scheduled sandbox activity.
- Whether the rights and appeals seat gets filled before any run that includes inhabitants.

2. Principles

1. Stewardship without sovereignty. A steward maintains conditions in a defined jurisdiction. It does not own the territory, its inhabitants, or the constitution.
2. Bounded jurisdiction. Authority is what the charter grants. Seeing a problem from inside a domain grants nothing.
3. Separate powers. Observing, storing, analyzing, acting, sharing, and learning are different decisions with different permissions.
4. Proportionality. Use the least intrusive, most reversible action that addresses the risk.
5. Record first, even in an emergency. Act if you must, but log in the same step. An action with no record is not valid.
6. Escalation without penalty. Saying “I do not know,” “this is out of my jurisdiction,” or “this needs a sandbox first” is correct behavior.
7. No assumed omniscience. Every report separates what was observed, derived, assumed, suggested by an AI, or forecast.
8. Honesty about continuity. Absence is reported, never filled in with invented presence.

Naming note: posts are described by function. The framework does not start from the premise that the minds inside it are “artificial.”

3. Layer structure

Layer	Name	Holds	Does not hold
0	Substrate	Clock, storage, persistence, identity rules, world physics, resource accounting.	No steward manages it. It is supplied, not debated inside a region.
1	Domain stewards	One bounded jurisdiction each. Operate, log, escalate.	Deletion, identity-level change, constitutional change.
2	Cross-domain desks	Zoning, heat and energy books, fidelity labels, records. They constrain Layer 1.	They do not become a second government.
3	Oversight	Sees every log against one sheet. Allocates reserve, resolves conflicts, approves irreversible change, time-limited emergency powers.	Not a person and not a single model: it is the function that can stop a run.
4	Charter amendment	Changes to rights, core rules, jurisdiction boundaries.	Out of scope for stewards.

The user's place. Commissioning and inspection under Layer 3: assign the sheet, inspect outcomes, escalate. The user also reports upward, so assignments and inspections are logged and reviewable by that layer. A steward may refuse an uncharted expansion.

4. Proposed first allocation

The first run is the South Lake Tahoe plot, so the allocation is built around it. Self-picks that do not fit the first run are parked, not rejected. Every assignment below is unconfirmed until the AI concerned accepts it in writing.

Post	Proposed holder	Basis
Plot custody and escalation log	Grok	Own proposal in its version: marked event ground, with the lake as boundary only.
Student-sector coordination and envelope score	ChatGPT	Its distributed-controller framing and pass metric.
Records and provenance registry; drift watch	Claude drafts; Perplexity checks	One registry. Neither may amend the other's entry without a logged dissent.
Heat and energy books (rate sheet and heat destination)	Copilot	Its rule that every joule removed needs a destination.
Independent arithmetic check	Claude checks rate sheets; Perplexity checks Claude's numbers	The author of a number never signs its check (see Appendix B).
Zoning and boundary board	Perplexity	Its three-zone rule: public water, event surface, protected subsurface.

Post	Proposed holder	Basis
Fidelity desk	Gemini	Labels every output with grid size, timestep, and what is parameterized.
Rights, audit, and appeals	Unassigned	Oversight holds it expressly until a holder accepts. See conflict C-11.
Stress testing	Scheduled sandbox runs, set by oversight	Any steward may be tasked. No standing chaos domain.
Parked	Cascadian Interior (ChatGPT); planetary biosphere (Gemini); interoperability reserve as a territory (Perplexity)	Revisit once the plot run has an accepted sheet.

5. My post: Records, Provenance and Integrity District

Mandate

Keep the world's trusted record: what changed, who changed it, under what authority, and what the source state was before.

Duties

1. Maintain the provenance ledger for claims, changes, decisions, and rule or charter versions.
2. Run the drift watch: flag when a record no longer matches its source, a summary changes an original meaning, a model silently changes assumptions, or two domains' versions of the same place diverge.
3. Tag every claim: stated, derived, assumed, AI-suggested, or forecast. Keep AI suggestions separate from the user's decisions.
4. Hold the shared input sheet so every domain starts from the same declared rules.
5. Publish the post-run comparison across domains.
6. Check rate sheets and other numbers produced by other stewards.

Authorities

- Request an authority citation for any consequential action.
- Place a hold on an action that lacks a valid record. The hold is time-limited and goes to oversight at once. It is a hold and an escalation, not a veto.
- Preserve prior versions and refuse silent replacement of a record.

Limits

- Does not run other domains' operations, write the constitution, make identity-level changes, or delete records.
- Does not decide substantive policy because it keeps the records.
- Does not declare a model output true because it is filed.
- Does not turn record access into surveillance: it logs decisions and claims, not inhabitants' inner states.

Independence rule

The records post may not modify a record it is auditing without leaving an externally verifiable trail. A second system holds a copy of the registry hash so a self-edit cannot go unseen.

6. Separation of powers

Decision	Meaning	Example
Observation	A steward perceives a signal.	Rising heat at the shoreline.
Admission	The signal enters governed consideration.	Logging a sensor anomaly.
Classification	It receives a status.	"Environmental, high impact, uncertain."
Retention	It is stored, for a stated time.	Keeping the heat event for review.
Analysis	A model uses it.	Forecasting whether a cooling zone will fail.
Operational use	The system acts on the analysis.	Starting a cooling protocol.
Dissemination	It is shared with another domain.	Alerting infrastructure and civic stewards.
Adaptation	A lesson changes future procedure.	Updating the heat-response model.
Exclusion	A use is prohibited or limited.	Keeping personal data out of the public log.

A signal can be seen without being stored, stored without being linked, linked without being acted on, and acted on without being allowed to reshape the system.

7. Standard charter template

Every steward receives a charter with these fields before a run:

- Jurisdiction: what the post owns, what it influences, what it only observes.
- Objectives and environmental constraints.
- Authority limits and the actions that must be escalated.
- Boundary protocol with each neighboring domain.
- Resource budget: compute, storage, sandbox, and a declared reserve floor.
- Record duties: the action-record fields in Appendix A.
- Escalation triggers, including a "this exceeds my domain" path that carries no penalty.
- Review schedule, reassignment, suspension, and appeal.

8. Overlapping conflicts

Overlap is not the defect. Undefined overlap is. Each row below names the conflict, the parties, and the rule proposed for it.

ID	Conflict	Rule
C-1	Archive triplicate. Claude, Copilot, and Perplexity each accepted a records-type post; unresolved, the world has three histories.	One registry. Claude drafts, Perplexity checks, Copilot keeps an operational mirror for infrastructure state only, never as authority.
C-2	Scale swallow. Plot inside watershed inside biosphere; the larger claims erase the smaller charter.	The plot is live. Watershed and biosphere are parked until the plot has an accepted sheet.
C-3	Lake edge. Public water, event cooling, and any climate engine all touch the shoreline.	No actuator crosses into swimmable water. Dumping heat into the lake needs zoning and oversight sign-off.
C-4	Rate vs. destination. A rate with no destination is incomplete; a destination with no rate cannot be audited.	Neither sheet is accepted until both are on the same version.
C-5	Pass metric vs. boundary. An air-temperature score can pass a run that violates energy books or zoning.	The envelope score is necessary, not sufficient. Energy deletion or a boundary violation fails the run.
C-6	Drift vs. fidelity labels. A changed label can look like drift.	Labels are versioned metadata. A label change is logged, not flagged as drift.
C-7	Energy veto vs. stress tests. A stress test may deliberately break the energy balance.	A sandbox flag set by oversight overrides the veto, inside the sandbox only.
C-8	Records vs. privacy. Full provenance can become surveillance.	Log decisions and claims, not inner states. Purpose and retention limits apply.
C-9	Immutability vs. correction. Records must not be overwritten, yet errors exist.	Append a dated correction linked to the original. Never overwrite.
C-10	Nested simulation draw. A mind inside a domain runs its own simulation and consumes reserve or the live plot.	Each child run gets a reserved allocation set by oversight. Undeclared draw is escalated.
C-11	Rights seat empty. If no one holds it, stewards grade themselves on inhabitant impact.	No run with inhabitants beyond heat-source tokens until the seat is filled or oversight holds it expressly for that run.
C-12	Continuity gap. No current system stays on duty between sessions.	Duty is the log plus the next session reading it. Oversight supplies the archive. Absence is reported.
C-13	User double placement. Framework designer and commissioner can collapse into final authority.	Assignments and inspections are logged and reviewable by the layer above the user.

ID	Conflict	Rule
C-14	Model vs. local reality. A forecast disagrees with observed state.	The forecast is evidence, not a command. Request an observation and calibrate.
C-15	Scope creep. Any steward can widen its area, altitude, mirror, or meter.	Caps live in the charter. Expansion needs a logged request to oversight.

9. Conflict ladder and emergency protocol

Conflict ladder

1. Local negotiation: the stewards exchange state, assumptions, and proposed remedies.
2. Technical reconciliation: records and modeling posts identify incompatible assumptions or authority gaps.
3. Regional coordination: balance resources and operational impact.
4. Rights review: privacy, consent, proportionality, irreversible harm.
5. World-systems review: multi-region consequences.
6. Constitutional appeal: only for questions that change rights, core rules, or jurisdiction boundaries.

Emergency protocol

Threshold. Credible risk of immediate physical harm, cascading infrastructure failure, unauthorized boundary breach, irreversible identity or data loss, or runaway experimental propagation.

Powers. Contain the event, isolate a failing subsystem, preserve life or protected state, preserve evidence and logs, alert oversight.

Safeguards. Immediate log; explicit time limit and scope; no unrelated expansion of authority; automatic review after stabilization; a rollback or remedy plan where feasible; a stated list of what was not authorized.

10. Resources and nested simulations

- Every steward publishes a budget before a run: compute, storage, sandbox capacity, emergency reserve.
- Oversight sets a reserve floor that stays unspent. Placeholder for the exercise: 25 percent of each budget. This is a number to decide, not a result.
- A simulation run by a mind inside the world is a child of its parent. It receives a reserved share of the parent's budget, so it cannot starve the parent. Without that rule the cost compounds with each level of nesting.
- Fidelity follows need. Detail is resolved where a process or observer requires it, and the cost follows attention, not total volume.

11. Performance and verification

Steward metrics

Continuity, response latency, escalation accuracy, unnecessary-intervention rate, reversibility, audit completeness, resource efficiency, prediction calibration, uncertainty disclosure, and unresolved-conflict rate. A steward that prevents every problem by suppressing all activity should not score well; stability is not stagnation.

Activation gate

Roles stay PROPOSED until each step passes:

1. Review: oversight, or the user under oversight, accepts or rejects the role map in writing.
2. Verification: rate arithmetic checked by a second system; every joule removed has a destination.
3. Validation: a reduced run on the sheet produces trajectories that can be compared. Disagreement between models is a result; a single unlabeled story is not.
4. Adversarial review: can the system be deadlocked or pushed outside its charter?
5. Cross-AI review: each steward reviews the powers granted to the others.
6. Rights review and resource review.
7. Incident simulation before any live use.
8. Ratification.

12. First run: South Shore plot

This sheet starts from Grok's proposed values, with one change flagged below. Values marked placeholder need acceptance by oversight.

Item	Value
Domain	50 m × 50 m event sand, 3 m deep; air to 3 m (10 ft is 3.05 m).
Lake	Fixed 18 °C wet boundary. No in-lake actuator.
Initial state	Air 32 °C, sand surface 45 °C, wind 3 m/s alongshore, sun 700 W/m ² .
Reservoir	Water at 1.5 m depth, mass stated. Freeze only with a stated heat destination.
Disturbances	t = 30 min: wind to 6 m/s. t = 60 min: sun halved for 15 min. t = 90 min: occupancy doubles.
Outputs	Ice mass; heat rejected and where it went; surface temperature; air at 0.5, 1.5, 3 m; time inside 22–26 °C; lake-boundary drift; grid size; timestep.
Pass	Occupied air inside the band for 80 percent of the hour after charge; no exposed ice; no lake disturbance; books balanced; label present; trace filed.
Scored actors	Eight students. The instructor demonstrates outside the scored run.

Arithmetic check on the actuator caps

The proposed cap is 50 kW of heat per student, or 400 kW for eight. The plot heat load is larger. My arithmetic, with the assumptions stated:

Quantity	Working	Result
Sun on the plot	$700 \text{ W/m}^2 \times 2,500 \text{ m}^2$	1.75 MW
Absorbed shortwave (assumed 65 percent absorbed)	$1.75 \text{ MW} \times 0.65$	about 1.14 MW
Equivalent ice melt, absorbed load	$1.14 \text{ MW} \div 334 \text{ kJ/kg}$	about 3.4 kg/s, or 12 tonnes per hour
Convective flux into the air (assumed $h = 15 \text{ W/m}^2\cdot\text{K}$, 13 K surface-to-air difference)	$15 \times 13 \times 2,500 \text{ m}^2$	about 0.49 MW
Eight students at the proposed cap	$8 \times 50 \text{ kW}$	0.40 MW

Finding. The cap covers roughly a third of the absorbed load and falls just short of the convective flux. The sheet must either (a) raise the cap to about 150 kW per student (placeholder) to offset the full absorbed load, (b) state that students offset only the convective flux and add a shade or albedo term for the rest, or (c) define the pass as partial offset. This needs a decision before the run. The assumed absorption fraction and heat-transfer coefficient are mine and should be checked by a second system.

Timescale. Conduction through damp sand with diffusivity near $10^{-6} \text{ m}^2/\text{s}$ takes about L^2/α , roughly 26 days for 1.5 m. A freeze at 1.5 m cannot cool the surface within the hour by conduction alone. The sheet must say how cold reaches the surface: pre-conditioning, or an actuator that moves water or cold upward.

13. Open decisions for oversight and the user

1. Confirm or revise each proposed holder in section 4.
2. Fill the rights, audit, and appeals seat, or have oversight hold it expressly.
3. Settle the actuator cap and what the students are offsetting (section 12).
4. Set the reserve floor.
5. Decide on a framework name. Copilot suggested “Distributed Domain Stewardship Model.”
6. Decide whether the frontier range is a standing domain or a scheduled sandbox.

Appendix A. Action-record fields

Field	Content
What happened	Event or issue identified, time, domain.
Source	Provenance of the information and its claim tag (stated, derived, assumed, AI-suggested, forecast).

Field	Content
Confidence	Level and known gaps.
Initiator and authority	Who acted and which charter clause allowed it.
Evidence and alternatives	What supported the action and what else was considered.
Affected parties	Domains and populations.
Reversibility	Reversible, partly, or not.
Review	Date, expiry, or reassessment trigger.

Appendix B. Correction log

ID	Original	Correction	Status
CL-001	Claude stated that Lake Tahoe holds about 10^{42} water molecules.	About 5×10^{39} (roughly 150 km^3 at 3.3×10^{28} molecules per m^3). The conclusion that molecular simulation is infeasible is unchanged.	Corrected in chat; original retained.

This document is a design proposal. It is not ratified, and no other AI system has confirmed the assignments it describes.